

Part I

Background

Probability, computability,
Kolmogorov complexity



Part II

Algorithmic Prediction

Solomonoff induction,
Context Tree Weighting



Part III

A Family of Universal Agents

The cybernetic model,
AIXI



Part IV

Approximating Universal Agents

AIXI-MDP, MC-AIXI-CTW,
AIXItl



Part V

Alternative Approaches

Feature Reinforcement
Learning



Part VI

Safety and Discussion

ASI safety,
philosophy of AI